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Spatio-temporal GIS Development to Aid Multiple-use Forest Management

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Abstract

This research investigates the potential of a dedicated Geographic Information System (GIS) to provide spatiotemporal data visualisation and analysis to aid multiple-use forest management in the Kingston State Forest, near Manjimup, Western Australia. The goals of this research are to develop and use sophisticated data evaluation techniques to assist in understanding the ecology of the western ringtail possum and to evaluate the impact of current timber harvesting practices on possum populations to ensure the jarrah forests continue to support them.

Data used included digital topographic data as well as spatial and temporal fauna observation data. Visual representation of interactive queries offered a conceptual understanding of the spatial structure and temporal aspects of the data, showing that possum behaviour differed between harvested and non-harvested areas within the research site. Furthermore, the dedicated GIS proved its utility as a management and research tool that is readily transferable and accessible to multiple-use land managers and research scientists, offering a generic environment for subsequent fauna habitat research. The focus of this paper is the development of the dedicated GIS environment.

Introduction

Human interaction with the surrounding environment invariably leads to changes in ecosystems and their ecological balance. Fauna management embraces the concept of understanding the environment in an attempt to better protect native species. One major threat to the native fauna is the reduction of suitable habitat.

The south-west native forests of Western Australia are currently managed for multiple uses, including conservation, recreation, timber production, and water resources. While these forests are significantly less altered than many other Australian ecosystems, they have been, and continue to be, exposed to human-induced modifications such as farming and timber harvesting (Hobbs, 1996).

In 1994, Western Australia's Department of Conservation and Land Management (CALM) initiated the Kingston

Project in recognition of the need for ongoing research into the impact of timber harvesting and associated activities on the fauna and flora of these native forest ecosystems (Morris *et al.*, 2000). Individual studies examined the responses of the vegetation (floristics and structure), abundance of hollow-bearing trees, small terrestrial vertebrates, medium-sized mammals, and birds over a five to 10 year period (Burrows *et al.*, 1994; Morris *et al.*, 1996; CALM, 1997b). Several species of medium sized mammals were tracked using radio-telemetry, providing an opportunity to develop and evaluate the use of a GIS for spatio-temporal data analysis to supplement the standard statistical analyses. The radio-telemetry data being collected on the Western Ringtail Possum (*Pseudocheirus occidentalis*) were chosen for this research, both in recognition of the species' potential for decline due to harvest activities and the volume of spatial data being collected on this species.

A dedicated GIS was developed and used as a platform for static and animated data representation, accompanied by the development of specific interactive queries offering a unique insight into various spatial and temporal relationships. The dedicated GIS was developed as a generic platform suitable for alternate data sets from within the Kingston Project, as well as being adaptable to the needs of subsequent fauna research.

Site Description

Kingston Forest Block

The western ringtail possum study has been carried out within the Kingston Forest block, 26km north-east of Manjimup in Western Australia's southwest. The Kingston block lies adjacent to the Winneup and Warrup blocks (Figure 1). The study area is approximately 5,000ha, comprising open forest dominated by jarrah (*Eucalyptus marginata*) and marri (*Corymbia calophylla*) in the intermediate rainfall zone (Morris *et al.*, 2000). In the 1920's, the area was selectively harvested for high quality jarrah, and again between 1940 and 1976 (Herberle, 1997). Prescribed fuel reduction burns on a seven to 10 year rotation have also been implemented since the 1960's (Morris *et al.*, 1996).

Harvesting Techniques

Road construction within the study area was conducted during December 1996 and January 1997 to create access for harvest operations. Harvesting operations were carried out from March through to August 1997 and completed during the November 1997 to January 1998 period (Wayne *et al.*, 2000).

Jarrah Stand Improvement (JSI) operations were carried out following the completion of tree falling and extraction operations. JSI within gap release areas involved CALM operators preparing the area for maximum potential

regeneration by removing most of the remaining standing trees (non-merchantable timber), excluding primary and potential habitat trees. This operation reduced suppression of the saplings and lignotubers, thus encouraging rapid timber regeneration. The final step was a post-harvest silvicultural burn on 9th November 1998 (Wayne *et al.*, 2000).

Fox Control

In June 1992, prior to the planning of the Kingston Project, fox control using dried meat baits was initiated in the area. This exercise was part of the operational expansion of CALM's fox control programs in multiple-use forests, known as Foxglove and Western Shield (Bailey, 1996). As anticipated, the fox population within the area was reduced and an increase in the abundance of medium-sized native mammals was detected, enabling the impact of harvesting to become the focus of research in the region (Morris *et al.*, 2000).

Research Objectives

The aim of this research was to develop a customised, dedicated GIS environment for use in multiple-use forest management. In addition, this research endeavoured to present a more comprehensive account of the ecology of the western ringtail possum and, in turn, achieve a better understanding of the impact of timber harvesting on western ringtail possums within the jarrah forests of Western Australia. Further, the dedicated GIS was intended as an aid for improved communication and collaborative decision making among various experts involved with multiple-use forest management.

Research Data

Field Data

The field data used for this GIS research were collected as part of the Kingston Study (Wayne *et al.*, 2000). The



Figure 1a Site location map within Australia.



Figure 1b Site location map, including treatment and control areas of the Kingston Study.

specific aims of the Western Ringtail Possum research were to investigate survivorship, habitat use and home ranges of the western ringtail possum with respect to harvest impacts (Wayne *et al.*, 2000).

Twenty-nine possums were fitted with collars containing mortality sensitive radio-transmitters (Biotrack two stage transmitters) (Wayne *et al.*, 2000). Individual possums were subsequently tracked for regular monitoring and gathering of data with respect to habitat use, movement, and mortality (Wayne *et al.*, 2000; Morris *et al.*, 1996).

Data were collected for three months pre-harvest and were continued post-harvest until all experimental possums had either died or radio-signal contact was lost. Day fixes of radio-collared possums were recorded to identify the day refuges (location and characteristics) being used by individuals and to help determine the characteristics of these refuge sites and the frequency of their use. The possums were usually surveyed four days per week (CALM, 1997a). Where possible, a minimum of 20 day refuge fixes were obtained per possum, both pre- and post-harvest, in order to provide a reliable measure of refuge range. Physical parameters and breeding status of western ringtail possum individuals were updated each time the radio collars were replaced, approximately every six months. Disturbance data were collected at the commencement of harvest operations, relative to the known range of each experimental animal. The bodies of dead animals were examined to determine the cause of death (CALM, 1997a).

When radio-collared western ringtail possums were located during daytime radio-tracking surveys, details regarding their refuge location, refuge characteristics, surrounding vegetation, and landscape were recorded. Refuge characteristics included refuge type, species, senescence, height, diameter at breast height, and location.

During the data collection phase, CALMScience staff entered field data into a custom designed relational database (Microsoft Access). Review of the field data stored in the relational database was performed to validate the data set. Spatial examination of the data during the validating process proved very useful in identifying data discrepancies. A number of data discrepancies required further field verification, while, in a few rare cases, the data in question were removed from the database as there was no reliable method of data verification.

Spatial Data

Possum refuge locations were collected for each newly observed refuge using a standard compass and chain relative to reference points with known GPS coordinates that were pegged along logging and access roads. These *in situ* data were then converted to coordinates (Eastings and Northings) and added to the relational database.

Spatial data accuracy varied due to many contributing factors, including the lack of accountability for the natural topography and some approximated distances (due to dense vegetation). In theory, all refuges were spatially located

with respect to a known reference point. In practice however, many refuges were located via a chain of other refuges before they were eventually anchored to a known GPS reference point. Therefore, errors in the spatial location of one refuge were compounded when used as a reference for the position of another refuge.

These errors were reduced using a set of preliminary maps of refuge locations, a list of refuges with multiple spatial references, and a list of refuges tied indirectly to the GPS reference points. These maps were examined by CALMScience staff, and corrected where discrepancies were observed. In some cases, further field reconnaissance was carried out until all known errors were amended and propagation of errors minimised.

Base Datasets

Existing environmental data relevant to the multiple uses of the Kingston Forest were supplied by the GIS Division of the Information Management Branch (IMB) of CALM. These base datasets were supplied digitally, clipped to the study area, and imported into the GIS. The datasets include contour lines, hydrography, vegetation, state forest boundaries, plantations, silviculture, roads, and forest blocks.

GIS Research Methodology

The methodology adopted for this research consisted of four stages: a requirement study, prototyping, construction, and testing (Figure 2). The purpose of the requirement study was to identify the expectations of the dedicated GIS, consider design issues and future applications, and finally assess hardware and software needs. Prototyping involved the development of efficient data flow between software applications, the design and development of interactive queries, and the structure and layout of the user interface. Construction of the GIS environment involved data preparation, extensive scripting for analysis tools and interactive interface design, and exploratory animation development. The testing stage was associated with both the prototyping and construction stages, and consisted of three levels: unit testing, system testing, and user testing. Unit testing ensured that individual scripts ran successfully, while system testing ensured complete script compatibility when implemented as a whole. User testing entailed the operation of the dedicated GIS as it was intended, for analysis of the impact of timber harvesting on the western ringtail possum. It also required the identification of any problems or restrictions in its design or execution so that final adjustments could be made. The final task of this research was to identify the limitations of the dedicated GIS.

Although the CALMScience staff involved in the western ringtail possum research had little understanding of how to develop and implement their GIS ideas, it was possible to construct a brief yet comprehensive list of criteria, based upon the user needs analysis:

- (a) Creation of an intuitive, interactive, yet flexible GIS environment;

- (b) Provision of visual alternatives such as maps, graphs, and tables;
- (c) Design of specific fauna movement and harvesting impact analysis tools;
- (d) Focus on spatial and temporal aspects of data sets; and
- (e) Generic design to support diversity in future applications.

Dedicated GIS Construction

Analysis Tools

The custom analysis tools were developed within the GIS program ArcView using Avenue, ESRI's proprietary development language. The objectives were to satisfy questions such as: where? when? why? how much? and how often? A standard approach was developed with individual variations to address specific queries.

Various refuge queries were implemented for highlighting refuge types, defining refuge popularity, and exploring the characteristics of each refuge. Identification of the influential parameters of the refuge environment was provided through queries regarding landscape position, vegetation height, percentage canopy cover, tree species within each storey, and vegetation litter. Queries were also developed to determine the distribution, quantity, and type of refuges inhabited by male, female, and individual possums.

Refuge range, core range, and range overlap queries were based on the spatial locations of observed refuges using the minimum convex polygon equation (Hooge and Eichenlaub, 1998). Refuge range queries offered monthly, seasonal, and annual assessment as well as temporal assessment with respect to the harvesting period. Refuge frequency was also combined with home (refuge) range to emphasise the lack of homogeneity within each range by identifying core refuges (Inions, 1985).

Graphical User Interface (GUI) Design

McDonnell and Kemp (1995) describe the GUI as the junction or link between user and software. A simple and straightforward GUI design was considered the best suited for non-expert users. The development scenario was based around an introductory interface, an analysis interface, an interchangeable map base, and coordinated interactive query mechanisms (Figure 3). The concept of an introductory interface offered a preface to the GIS system, granting access to background documentation as well as the analysis interface. The standard interface for the GIS was used as the basis for development of the introductory interface, and a clone was created for the independent development of the analysis interface.

The introductory prototype was primarily designed to provide topical focus and offer background information associated with the analysis project. It soon became evident that the introductory interface design was cluttered and would not accommodate expansion for



Figure 2 Custom GIS data, interface, analysis, and output methodology.

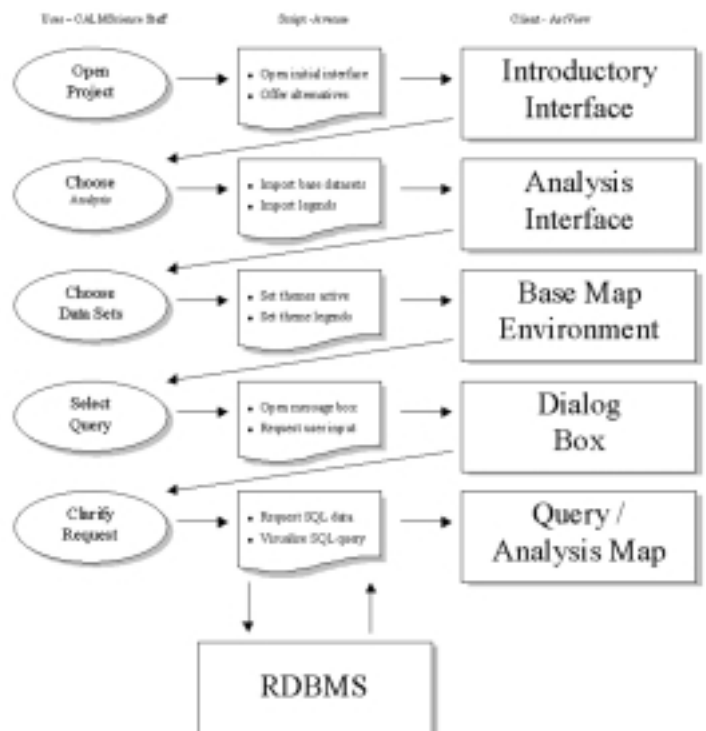


Figure 3 Diagram illustrating the automated query structure built into the custom GIS user interface.

alternative faunal data sets. The decision was made to expand the introductory interface to embrace a number of associated pages (Figure 4). The revised scenario effectively accommodated background information and analysis options, and demonstrated the ease with which the GIS could embrace future integration of data from the Kingston Project's medium-sized mammal studies.

The structure of the introductory interface was designed to feel comfortable and familiar, with the layout standardised among pages. All standard ArcView menus, tools and buttons were removed and the title 'Harvesting & Habitat Analysis' was assigned to replace the standard title. A graphics extension was activated to allow images to be imported, and interactive buttons were grouped according to function and set to a standard size to accommodate all text options, offering access to various background documents as well as the analysis interface. The background material was accessed through hotlinks to either word-processing documents or relevant web sites. As a gateway to the chosen analysis, the individual fauna pages were designed simply, with an image and title to verify the choice of fauna analysis made by the user. The analysis interface was a composite environment, including the map base and an analysis component. All standard menus, buttons and tools were considered for their relevance to the project and were either kept, modified, or removed. The analysis component consisted of a customised set of menus offering a selection of queries. Menus were selected as the principal interactive query method since their design was both the simplest and most familiar for novice users. Queries were clustered and assigned to custom menus according to refuge, possum, and silviculture based categories.

The map base was created using base data sets and accompanying legends (Figure 5). A script automatically imported the base data sets and listed them within the legend. The choice of active data sets was made by the user, and only active data sets were represented in the map base and clarified within the legend. The choice of data sets was flexible and could be modified via the interactive legend. The analysis interface offered an uncomplicated layout, designed specifically for non-expert users.

Design of the hardcopy map layout was left until all other aspects of design had been completed. As requested by CALMScience staff, each layout design featured an inset map in the top right corner to highlight the location of the refuges in relation to the greater Kingston area (Figure 6). The main maps containing the current queries occupied the remaining space, including relevant legends, north points, scales, and titles. Various hardcopy layouts were provided to accommodate different analysis scenarios, including multiple maps, A4 and A3 sizes, and portrait or landscape dimensions.

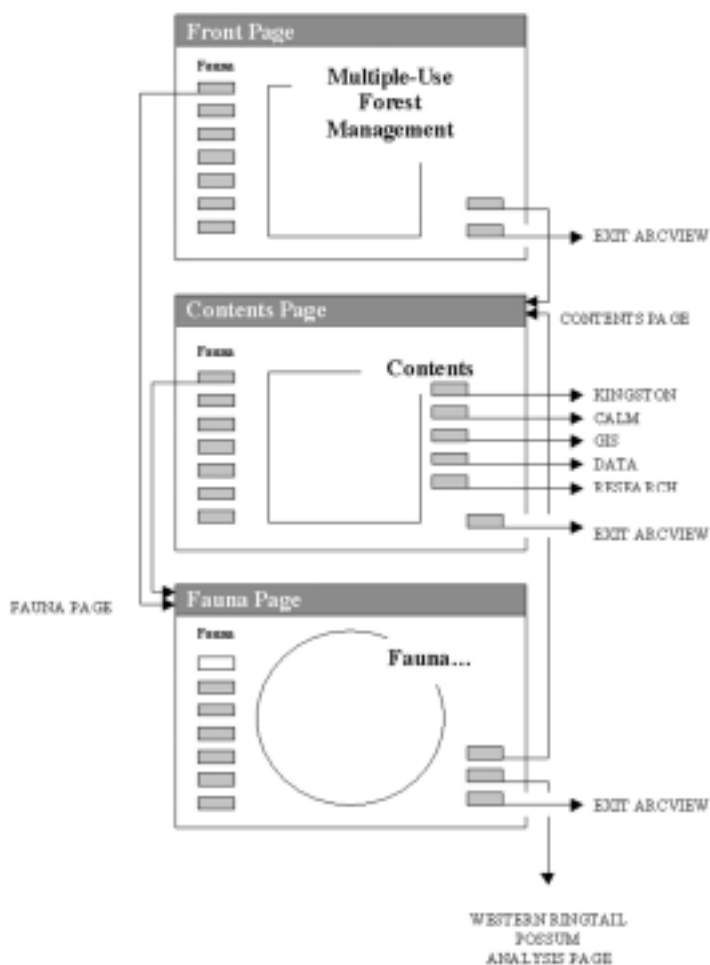


Figure 4 Simplified structure of the introductory interface illustrating and connecting the front page, contents page, and fauna page. Note that a user leaving the fauna page would either exit ArcView or move on to the possum analysis page.

SQL Development

Each menu selection and subsequent dialog box selection ultimately activated a script that sent a request to the database. For every possible query, the most suitable data combinations were determined and an SQL (Structured Query Language) request was constructed and embedded within the appropriate script.

There were three parts to the composition of each SQL request: data selection, table selection, and conditions of the selection. The conditions were used to tailor each request to individual query requirements. If the conditions listed within the SQL request were not met within the database, then some data would be forfeited in the request process. As some fields within the database tables were occasionally incomplete, SQL requests were constructed carefully to avoid unattainable conditions, thereby maximising the quantity of data returned by the request.

Script Construction

Numerous scripts were developed directly from the prototyping process, where the data flow concept and the development scenario laid out all functional requirements. The majority of scripts were directly associated with a particular page within the introductory

interface or with a particular menu or dialog within the analysis interface (Figure 7). A few standard GIS software scripts were also modified to customise the opening, closing and saving procedures for the project, along with ancillary scripts written to modify the automated legend.

The more complex scripts were associated with dialog box functionality. Individual scripts were attached to each object within each dialog box, dictating their responses.

The main script was attached to the OK button and carried out the major query functions. Once activated, the script dispensed with the dialog box, completed the SQL request with the database, carried out any necessary calculations, set the appropriate theme and legend parameters, and created a map layer within the analysis interface. The script also removed any unnecessary data tables and scaled the display with respect to the query. The most significant differences

among queries were the composition of the SQL request and the subsequent theme building process.

Similar structuring of scripts was used for the refuge frequency and movement queries. The refuge type and landscape position dialog boxes and queries were expanded to include the option of single refuge types or single landscape positions, respectively. The vegetation cover, refuge range, range overlap, and temporal queries required slightly varied designs, yet they remained similar in structure and functionality. The scripting process formed the basis of the dedicated GIS and was a complex and lengthy procedure involving the development of more than four hundred related scripts. See Figure 7 for an example of the SQL/script interface.

Animation Development

Animation technology offered the ability to observe possum movement and behaviour with respect to their changing environment. For example, the use of animation demonstrated the movement of the possums between refuges, illustrating the temporal relationships between refuges. Animation of refuge usage also highlighted the behavioural changes post-harvest, offering the chance to surmise the reasons for change in patterns. Independent animation software was investigated early on in the prototyping phase. However, during the research period an ArcView extension, Tracking Analyst, became available, providing the freedom to develop the animation as initially conceptualised.

An animation dialog box was designed with options for movement of possums by monthly, seasonal, annual, full study period, and pre- or post-harvest time frames (Figure 8). A supplementary dialog box allowed CALMScience staff to control the animation sequence. The animation sequence could be slowed down, accelerated, paused, repeated, or reversed. In effect, refuges were drawn as points in a temporal sequence based on field data collection dates - connected by lines which linked temporally adjacent refuges.

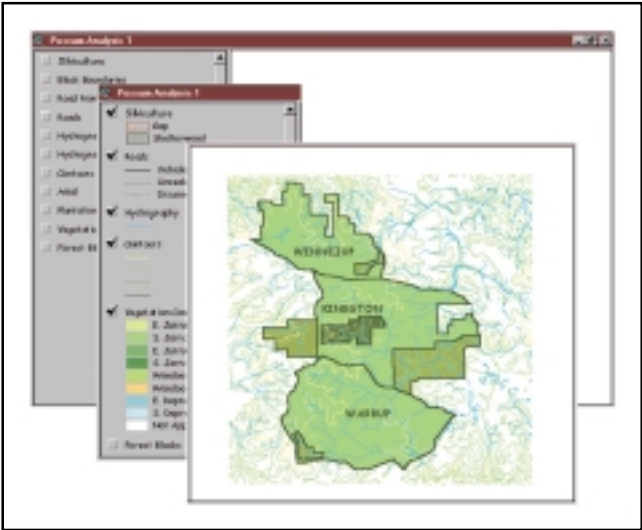


Figure 5 An example of the map base and automated legend of the analysis interface (base datasets).

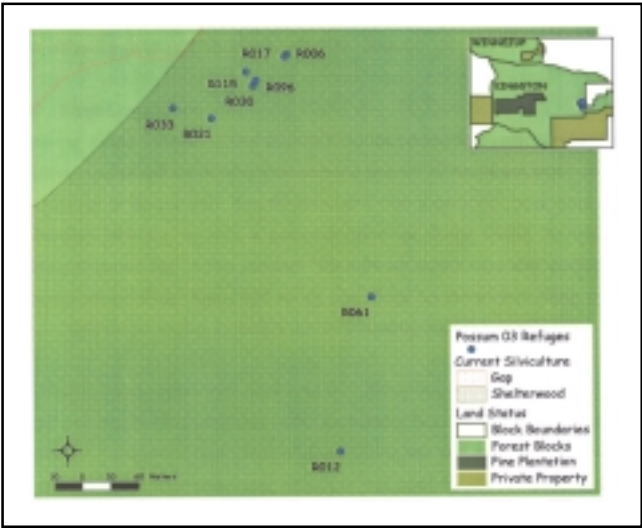


Figure 6 An example of a hardcopy layout with map insert showing the locations of possum 03's refuges.

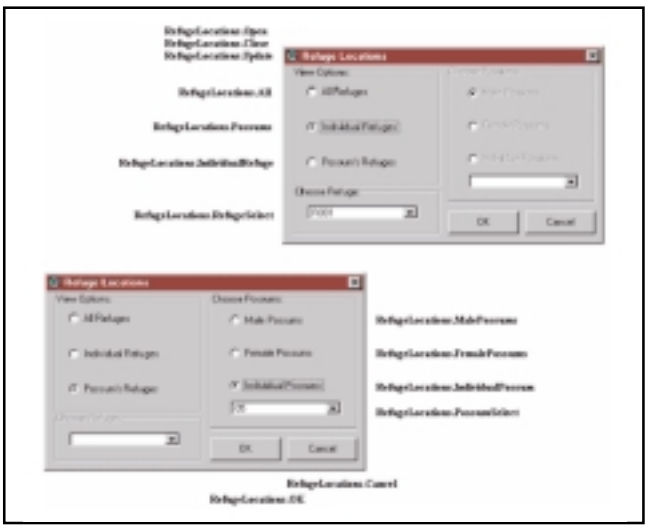


Figure 7 Examples of different custom SQL script queries built into the custom interface.

Analysis

General Analysis

A total of 2626 refuge occupancy records, relating to 413 different refuges, for 29 different possums were collected during the study. The possums were captured, collared, released, and then monitored in three geographically distinct areas: the eastern control area, the western control area and the treatment area (Figure 9). The treatment area, known as the Kingston 4 harvest coupe (K4), referred to the area that underwent harvest activity during the research period and was home to 17 experimental possums. The remaining 12 control possums occupied unharvested jarrah forest immediately to the east and west of K4.

Site Description

Of the nine vegetation complexes found in the Kingston Forest block, the experimental possums occupied the following vegetation complexes:

- Southern Jarrah Uplands: Tall open forest of marri and jarrah on uplands in perhumid and humid zones;
- Eastern Forest Depressions and Swamps: Tall open forest of jarrah and marri, woodland of flooded gum (*Eucalyptus rudis*) and swamp paperbark (*Melaleuca raphiophylla*), with some blackbutt (*Eucalyptus patens*) over river banksia (*Banksia seminuda*) and wonnich (*Callistachys lanceolata*) on valley floors in humid and subhumid zones;
- Eastern Jarrah Valley: Mosaic of open forest of jarrah and marri, woodland of wandoo with some powderbark wandoo (*Eucalyptus accendens*) and flooded gum on the deeper soils, ranging to closed heaths and lithic complex on shallow soils associated with granite on steep slopes of valleys in semiarid and arid zones; and
- Wandoo Valley: Woodland of flooded gum and blackbutt on footslopes on valley slopes in the subhumid zone (CALM, 1998c).

The eastern control area, situated slightly south-east of Yerraminnup Creek, had private property to the south, pine plantations to the north and various harvest coupes to the west. All four vegetation complexes were present within the eastern control area, yet wandoo valley was dominant. The western control area was situated on the southern boundary of a pine plantation and extended east to the edge of a private property boundary. Wandoo valley was again the primary vegetation complex, with a small area of southern jarrah uplands slightly to the south. Southern jarrah uplands and wandoo valley vegetation complexes were predominant in the treatment area (K4) west of Yerraminnup Creek, where shelterwood and gap release harvesting operations were implemented.

Tracking Period

The radiotelemetry database used in this GIS project is a large subset (data collected between February 1997 and June 1999) of the western ringtail possum radiotelemetry data collected by CALMScience staff. Of the 29 experimental



Figure 8 The user interface for animation query and control.



Figure 9 Overview map of possum refuge locations.

possums, Possums 01 to 18 (excluding Possum 07) inhabited the treatment area, Possums C01 to C07 occupied the western control area, and Possums C08 to C12, occupied the eastern control area. All experimental possums were fully independent adults when radio collared, with the exception of Possum 09, the 'at heel' offspring (sub-adult) of Possum 10.

The quantity of different refuges recorded as occupied by individual possums was influenced both by the length of their observation period and the number of observations made within that period. The tracking period for individual experimental possums began from their first day of capture until their time of death or the loss/failure of their radio collar. Observation periods ranged from five to 726 days, with a mean of 254 days. Occupied refuge records ranged from two to 226, with individual possums successfully radio-tracked to a refuge an average of 35 percent of the days within its observation period. See Figures 10 and 11 for examples of refuge use.

Refuge Types and Their Frequency of Use

The dedicated GIS was used to graphically show the

refuges used by possums based on their refuge type, their recorded characteristics, and to determine refuges on the basis of their frequency of use. During the study period, observations and records were made of 413 different refuges occupied by experimental possums. The number of different refuges used by individual possums ranged from two to 43, with an approximate average of 20 refuges occupied annually by each possum. Refuges used by the experimental animals were classified as: standing trees, balga bushes, above ground nests, forest debris, burrows, or hollow stumps (Wayne *et al.*, 2000).

Of the 413 refuges observed as occupied during the study, standing trees were the most commonly used (34 percent), followed by balga bushes, above ground nests and forest debris (25 percent, 21 percent and 19 percent respectively). Burrows and hollow stumps were rarely observed in use. While some possums appeared to favour one or two of the refuges, others roamed extensively and rarely used a refuge more than once or twice.

Refuge Range

The average refuge range for all experimental possums was 2.471ha (Tables 1 and 2), using the minimum convex polygon equation (Hooze and Eichenlaub, 1998) embedded in the home (refuge) range script. The average refuge ranges were 2.082ha for treatment possums, 1.101ha for western control possums, and 6.062ha for eastern control possums. For individual possums with more than 20 occupancy records, refuge ranges varied from 0.182ha to 11.077ha, with an average home range of 2.917ha. The average refuge range pre-harvest was 1.630ha, compared with 1.844ha post-harvest. Figures rose to 1.733ha pre-harvest and 2.803ha

post-harvest for treatment possums with 20 or more occupancy records.

The average refuge range for female treatment possums was 1.026ha, for female western control possums it was 1.260ha, and 1.553ha for female eastern control possums. Male treatment possums averaged 3.137ha, while the average refuge range for male western control possums was 1.048ha, and 9.069ha for male eastern control possums.

Some possums had only been detected in two different refuges during their observation periods; thus a refuge range calculation was unworkable. The smallest refuge range was established by Possum 16, a male treatment possum observed for 12 days in five different refuges over 0.178ha. Possum C11, a male possum in the eastern control area, established the largest refuge range of 11.077ha, observed over 236 days in 18 different refuges.

There were a few possums for which two refuge ranges were calculated based on their movement and establishment of new territories. Possum C01 established its first range in the western control area, which included nine different refuges and 0.302ha. After 185 days of observation, Possum C01 migrated to establish a second refuge range of 2.420ha, consisting of 34 different refuges over an observation period of 613 days. Possum 05, a female treatment possum, left its original refuge range of four refuges and 0.547ha after 22 days of observation. Possum 05 secured a second refuge range over 352 days of observation, covering 1.481ha and containing 15 refuges.

Core Range

Refuge range areas encompassed all parts of a possum's range despite the lack of uniform use of individual refuges.

Table 1 Details regarding refuge ranges of treatment possums - both pre and post harvest.

Possum	Sex	Refuges	Period (days)	Home Range (ha)	Pre-Harvest (ha)	Post-harvest (ha)
P01	M	9	68	1.983	1.983	N/A
P02	F	10	428	0.733	0.352	0.598
P03	M	9	168	3.759	3.759	0.026
P04	F	2	5	N/A	N/A	N/A
P05a	F	4	22	0.489	0.489	N/A
P05b	F	15	352	1.481	N/A	1.481
P06	M	14	124	5.372	5.372	0.018
P08	M	20	175	2.702	2.702	1.148
P09	M	8	143	1.367	0.49	0.971
P10	F	2	34	N/A	N/A	N/A
P11	M	13	184	2.588	1.86	6.209
P12	F	4	51	0.745	0.745	N/A
P13	F	6	81	1.185	1.186	N/A
P14	F	30	336	4.097	0.525	4.097
P15	F	10	323	0.446	0.093	0.438
P16	M	5	12	0.178	N/A	0.178
P17	M	35	391	7.509	N/A	7.509
P18	M	14	453	2.779	N/A	2.779

The concept of a core range is not new (Inions, 1985; Lindenmayer *et al.*, 1997) and is most often determined by the frequency with which refuges are occupied. For the purposes of this project, core ranges were based on a minimum of three refuges observed as occupied more than five times.

Of the 17 treatment possums, 11 utilised a core range of refuges. The average core range area for treatment possums was 0.824ha, varying from 0.102ha to 2.853ha. Four of the seven western control possums established core ranges varying from 0.176ha in size to 1.224ha, the average being 0.588ha. All five eastern control possums established core ranges varying from 0.029ha to 7.258ha, with 2.419ha the average. The core ranges generally occurred across the central part of the refuge range, with the extremes of the refuge range often being less extensively utilized (Figure 12).

Refuge and Range Overlap

Of the 413 refuges observed in use during the study period, 15 refuges were occupied by more than one radio-collared possum and one refuge was occupied by three possums. Of the 29 experimental possums, the refuge ranges of 13 possums experienced some overlap. The extent to which overlap occurred was in most cases minimal, ranging from 0.046ha to 1.293ha. Those that shared the majority of their refuge range had only established small refuge ranges to begin with, all under 1ha. It is important to note that not all western ringtail possums within a discrete area were radio-tracked. Therefore, these figures of home range overlap are not necessarily a true reflection of the extent of refuge range overlap. However, for the purposes of GIS research and development, these results demonstrate the capacity to accurately calculate more biologically meaningful information using appropriate data.

Possum Movement

Animation was implemented to more thoroughly examine the movement patterns and refuge range characteristics of individual possums within the treatment and control areas. Movement between two consecutively recorded locations was assumed to be a straight line and was unveiled dynamically at a daily rate, whether or not there were daily records.

The most revealing characteristic was the order in which refuges were occupied. The general trend appeared to be one of disorder, yet patterns did emerge when examined more closely on a case-by-case basis. The standard that came forth was of regular movement among the primary refuges in a criss-cross manner, interspersed with occasional sorties to new areas and, in some cases, eventual range movement to favour more recently used primary refuges.

Discussion

Impact Analysis

Having assessed some of the influencing aspects and the

resultant behaviour of individual possums within the study, observations with respect to timber harvesting impacts are possible. These observations are intended to aid in the management of multiple-use forests, with particular emphasis on the conservation of the western ringtail possum and its habitat.

Based on the subset of the data used during this study and on individual possums with 20 occupancy records or more, the average refuge range for treatment possums was 2.6ha; 1.7ha pre-harvest, compared with 2.8ha post-harvest. Increases in refuge range size were recorded post-harvest for both male and female possums, again based on 20 occupancy records or more. The average male refuge range was 2.7ha pre-harvest and 3.7ha post-harvest, while the female average refuge range increased dramatically from 0.6ha pre-harvest to 1.7ha post-harvest. One interpretation of these observations is that harvest activity impacted the behaviour of treatment possums, both male and female, sending them further afield to explore and establish suitable refuges, thus extending the size of their refuge ranges. It should be noted that this change may be a sampling artifact caused by the number of average pre- vs post-disturbance samples (33 and 60 observations, respectively).

The GIS maps and animations were also particularly useful in demonstrating that, for many of the radio-collared individuals, some refuges were used more extensively than others. Therefore, it appears that some refuges are of greater habitat value and, therefore, greater conservation value for western ringtail possums than others. Within forests subject to timber harvesting, all efforts should be made to protect those habitat elements that are of greatest conservation value. In conjunction with the spatial data, the physical attributes of those refuges that are used more extensively can be used to help identify what habitat elements most need to be protected and conserved.

Practical Applications

The development of a dedicated GIS was designed for immediate practical application using data from the western ringtail possum study. Consideration was also given to develop the GIS as a generic interface for further applications. Aspects such as data flow, configuration of the interface, query design and scripting were all developed so that minimal adjustments would be required to accommodate alternative and/or supplementary data sets.

Individual studies from the Kingston Project contained a number of suitable data sets, particularly the medium-sized fauna study begun in 1994. Similar to the western ringtail possum study, radio-tracked data from the common brushtail possum (*Trichosurus vulpecula*) study would be readily accommodated within the dedicated GIS. Analysis of other species such as the common brushtail possum would benefit greatly from access to visual and animated representations, and specific spatio-temporal query tools.

Trapping data were another form of spatial data collected in the medium-sized fauna study. Integration of

Table 2 Details regarding the refuge ranges of treatment possums.

Possum	Area	Sex	Refuges	Period (days)	Home Range (ha)
C01a	West Control	M	9	185	0.302
C01b	Migrated	M	34	613	2.42
C02	West Control	M	10	78	0.901
C03	West Control	F	8	91	0.182
C04	West Control	M	2	9	N/A
C05	West Control	M	24	455	1.177
C06	West Control	F	24	375	2.338
C07	West Control	M	30	726	1.486
C08	East Control	M	22	416	7.371
C09	East Control	F	7	365	0.584
C10	East Control	M	20	172	8.758
C11	East Control	M	18	236	11.077
C12	East Control	F	14	304	2.521

this form of data would also expect to be straightforward, however, extra scrutiny of the data structure and format would have to be carried out prior to implementation within the GIS.

GIS Evaluation

Visualisation was an essential component in the verification of data quality, both spatially and temporally. Visualisation and animation provided an insight into the general movement of possums, the size and character of refuge ranges, and of the varied use of primary and less frequently used refuges. Animation of possum movement also highlighted the change in movement and use of refuges post-harvest. These are all aspects of the data that would otherwise be difficult to assess purely from statistical data representations.

However, there were some aspects of the GIS environment that were limiting. Screen size placed limitations upon the scale of the geographic data, with loss of detail in small-scale maps and loss of overview in large-scale maps. Simultaneously preparing both hard-and-softcopy map presentations caused some reduction in quality, as choices that are best for on-screen presentation may not be the best options for printed maps (and vice versa).

As the temporal data were quite erratic, data animation was initially confusing. Refuge frequency data were added to all movement queries allowing the user to anticipate possible directional movement, and offering a perception of the time spent at each refuge. Movement with respect to harvest activity was restricted as polygon data could not be animated. Instead, colour was adopted for signifying movement with respect to harvest activity, one colour for occupation of a refuge pre-harvest and another for occupation post-harvest.

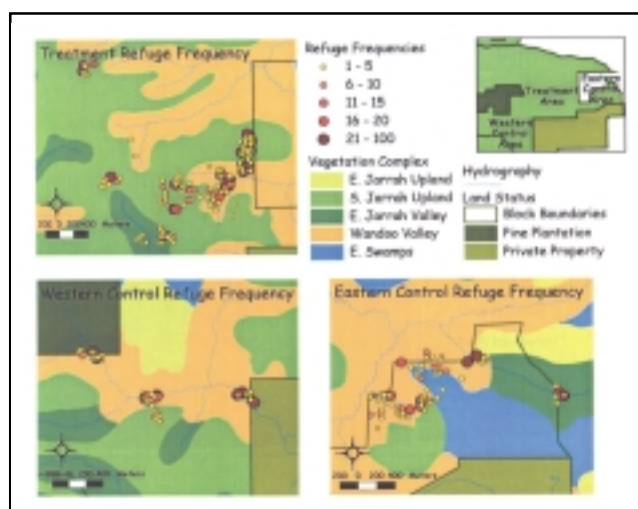


Figure 10 Illustration of frequency of refuge use in treatment and control areas.

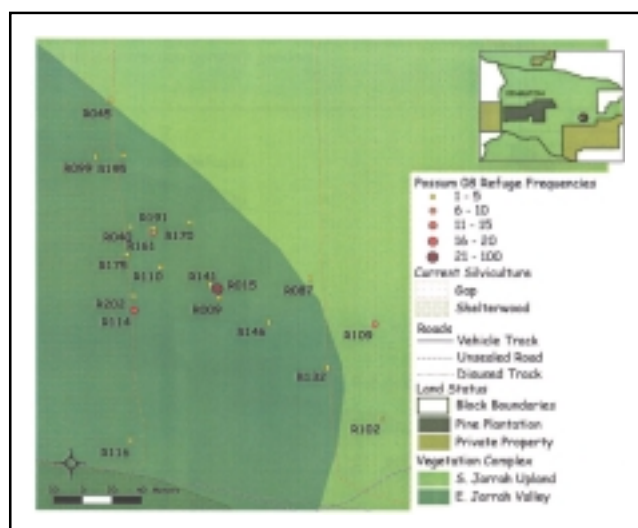


Figure 11 Illustration of the frequency of refuge use by possum 08.

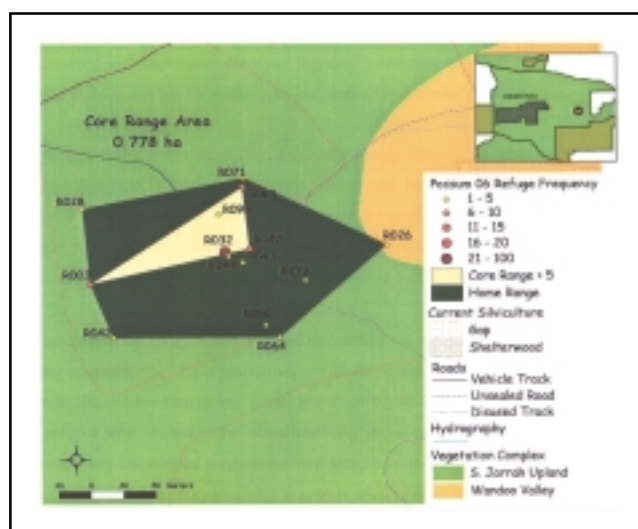


Figure 12 An illustration of refuge and core ranges for possum 06.

Significance

Although the western ringtail possum has been listed as an endangered species since 1950, relatively little has been discovered about the ecology of this arboreal marsupial. This research focused on the western ringtail possum and its behaviour, refuge requirements, and surrounding habitat in varied areas of the jarrah forest to better understand its needs within a multiple-use forest ecosystem. With this increased knowledge and understanding comes the chance to protect the western ringtail possum and its environment more effectively.

A significant aspect of the western ringtail possum GIS analysis was the recognition of a change in possum behaviour due to harvest activity. Recognition of the impact of timber harvesting and its repercussions within the forest ecosystem encourages further modification of harvesting practices and improved multiple-use forest management decisions.

The ecosystem is a dynamic environment that can be better understood when explored with respect to change over space and time. In this research, spatio-temporal analysis was implemented through the development of a dedicated GIS with specific environmental analysis tools accompanied by visualisation and animation capabilities. Without these assets, the depth of information contained within the data could not have been fully explored.

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Author's note: This research was carried out as part of Jane Lawley's MSc thesis in GIS at Curtin University of Technology (Perth, Western Australia). Dr. Hickey was her advisor and a Sr. Lecturer at Curtin at that time. Copies of the thesis (which include in-dept discussions and supporting graphics) (Lawley, 2000) can be obtained from the Department of Spatial Sciences at Curtin University (<http://www.cage.curtin.edu.au/spatial/>).